Automata

$\Sigma = \{a, b\}$

L_1 = Length 2

$\{aa, ab, ba, bb\}$

$\Downarrow$

finite

L_2 = starts with a

$\{a, ab, aa, aaa, aab,$
$aba, abb \ldots\ldots\ldots$
$abbbbba \ldots\ldots$
$aabababa \ldots\ldots\}$

$\Downarrow$

infinite

$\Downarrow$

Machine, Model

$\Downarrow$

Automata

baba

No

Language given $\Rightarrow$ Automata
L_2

Then decide string accepted or rejected.

Types of Automata

Grammar $\Rightarrow$

1. Finite Automata
2. Pushdown "
3. Linear Bound "
4. Turing machine

$\Leftarrow$ Language

Finite Automata

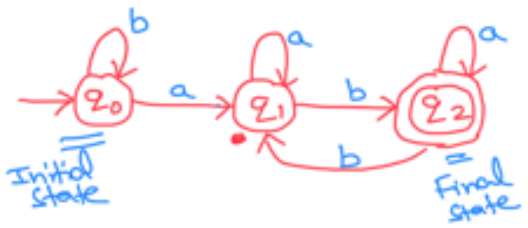
Finite Automata consists of 5 tuples.

$$M = (Q, \Sigma, \delta, q, F)$$

- Q → Set of states
- Σ → Alphabet
- δ → Transition function
- q → Initial state
- F → Set of final states

Check whether given strings accepted or not accepted ??

$\Sigma = \{a, b\}$



Strings

- ab ✓
- bab ✓
- bbaaba ✓
- abbb ✓
- aabab — Not accepted
- bbbaaab ✓
- aaa — Not accepted
- baaa — Not accepted
- bababa — Not accepted

abaaa Accepted ✓

$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a, b\}$

$q = \{q_0\}$

$F = \{q_2\}$

Transition function

$\delta(q_0, a) = \{q_1\}$
$\delta(q_0, b) = \{q_0\}$
$\delta(q_1, a) = \{q_1\}$
$\delta(q_1, b) = \{q_2\}$
$\delta(q_2, a) = \{q_2\}$
$\delta(q_2, b) = \{q_1\}$

Types of Finite Automata

1) Deterministic Finite Automata (DFA) ✓
2) Non-Deterministic Finite Automata (NFA) ✓

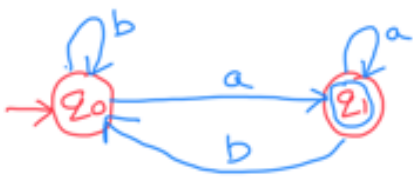
Deterministic Finite Automata (DFA)

Rule 1 – There should be transition from each state for every alphabet

Rule-2 – No two transition allowed from state for the same alphabet

Question-1 – Design a minimal state DFA in which every string ends with 'a'.
$\Sigma = \{a, b\}$

{ a, aa, ba, aaa, aba, baa, bba, aaaa ..... }



$M = \{Q, \Sigma, \delta, q, F\}$

$Q = \{q_0, q_1\}$ $\quad q = \{q_0\}$

$\Sigma = \{a, b\}$ $\quad F = \{q_1\}$

$\delta(q_0, a) = \{q_1\}$

$\delta(q_0, b) = \{q_0\}$

$\delta(q_1, a) = \{q_1\}$

$\delta(q_1, b) = \{q_0\}$

Transition Table

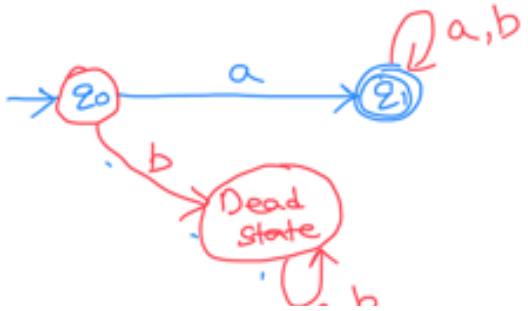
	a	b
→ q_0	q_1	q_0
* q_1	q_1	q_0

Question-2- Design a minimal state DFA in which every string starts with 'a'.

$\Sigma = \{a, b\}$

{ a, aa, ab, aaa, aab, aba, abb, aaaa ..... }

yes string starts with a



string which doesn't starts with a

No

$Q = \{q_0, q_1, \text{Dead state}\}$
$\Sigma = \{a, b\}$
$q = \{q_0\}$
$F = \{q_1\}$

$\delta(q_0, a) = \{q_1\}$
$\delta(q_0, b) = \{DS\}$
$\delta(q_1, a) = \{q_1\}$
$\delta(q_1, b) = \{q_1\}$
$\delta(DS, a) = \{DS\}$
$\delta(DS, b) = \{DS\}$

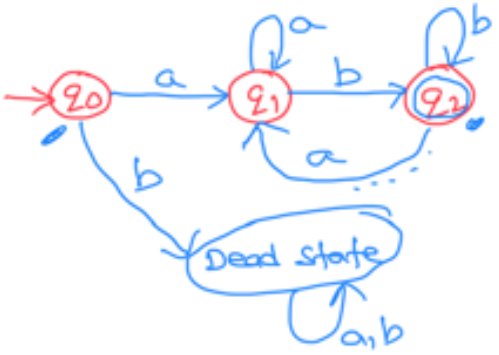
Transition Table

	a	b
q_0	q_1	DS
q_1	q_1	q_1
DS	DS	DS

Deterministic Finite Automata (DFA)

Question-3- Design a minimal state DFA in which every string starts with 'a' and ends with 'b'. $\Sigma = \{a, b\}$

$\{ab, aab, abb, aaab, abbb \ldots\}$



$M = (Q, \Sigma, \delta, q, F)$

$Q = \{q_0, q_1, q_2, DS\}$ $\quad$ $q = \{q_0\}$

$\Sigma = \{a, b\}$ $\quad$ $F = \{q_2\}$

$\delta(q_0, a) = \{q_1\}$

$\delta(q_0, b) = \{DS\}$

$\delta(q_1, a) = \{q_1\}$

$\delta(q_1, b) = \{q_2\}$

$\delta(q_2, a) = \{q_1\}$

$\delta(q_2, b) = \{q_2\}$

$\delta(DS, a) = \{DS\}$

$\delta(DS, b) = \{DS\}$

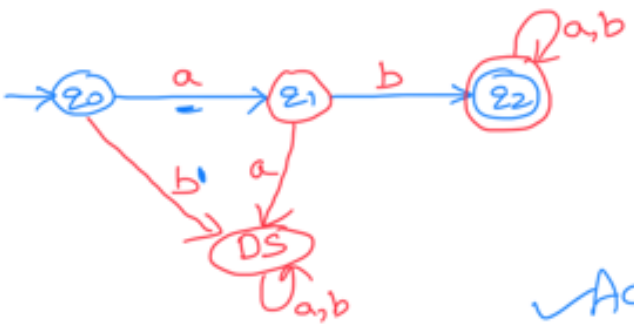
Transition Table

	a	b
→ q_0	$\{q_1\}$	$\{DS\}$
q_1	$\{q_1\}$	$\{q_2\}$
* q_2	$\{q_1\}$	$\{q_2\}$
DS	$\{DS\}$	$\{DS\}$

Question-4- Design a minimal state DFA in which every

String starts with 'ab'. $\Sigma = \{a, b\}$

{ab, aba, abb, abaa, abab - - - - - - - -}



Accept [ab aaabbba
ababab]

Not accept [aaababaa
bbabab
baaabb]

$M = (Q, \Sigma, \delta, q, F)$

$Q = \{q_0, q_1, q_2, DS\}$

$\Sigma = \{a, b\}$

$q = \{q_0\}$

$F = \{q_2\}$

$\delta(q_0, a) = \{q_1\}$
$\delta(q_0, b) = \{DS\}$
$\delta(q_1, a) = \{DS\}$
$\delta(q_1, b) = \{q_2\}$
$\delta(q_2, a) = \{q_2\}$
$\delta(q_2, b) = \{q_2\}$
$\delta(DS, a) = \{DS\}$
$\delta(DS, b) = \{DS\}$

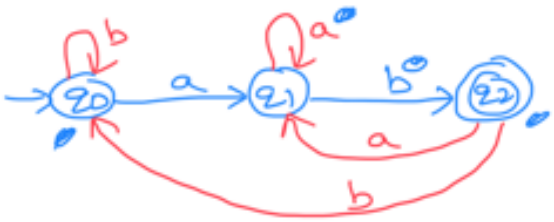
Transition Table

	a	b
→ q_0	$\{q_1\}$	$\{DS\}$

q_1	$\{DS\}$	$\{q_2\}$
$*q_2$	$\{q_2\}$	$\{q_2\}$
DS	$\{DS\}$	$\{DS\}$

Question-5- Design a minimal state DFA in which every string ends with 'ab'. $\Sigma = \{a, b\}$

$\{ab, aab, bab, aaab, abab \ldots\ldots$



Correct string [ababaab, bbabaab, abbbab] ⇒ Accept

Incorrect strings [bbbbaabb, aabbaa, babaaabb] ⇒ Not accept

$M = (Q, \Sigma, \delta, q, F)$

$Q = \{q_0, q_1, q_2\}$
$\Sigma = \{a, b\}$
$q = \{q_0\}$
$F = \{q_2\}$

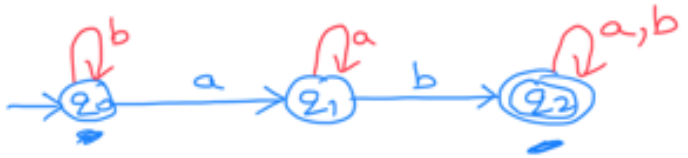
$\delta(q_0, a) = \{q_1\}$
$\delta(q_0, b) = \{q_0\}$
$\delta(q_1, a) = \{q_1\}$
$\delta(q_1, b) = \{q_2\}$
$\delta(q_2, a) = \{q_1\}$
$\delta(q_2, b) = \{q_0\}$

Transition Table

	a	b
→ q_0	$\{q_1\}$	$\{q_0\}$
q_1	$\{q_1\}$	$\{q_2\}$
* q_2	$\{q_1\}$	$\{q_0\}$

Question-6- Design a minimal state DFA in which every string contains the substring 'ab'.
$\Sigma = \{a, b\}$

{ab, aab, aba, abb, aaab, aaba, abaa .....}



Correct string [aaaabbaaba baaabaab] ⇒ Should Accepted

Incorrect string [aaaaaa bbbbbaaa baaa] ⇒ Should not accepted

$M(Q, \Sigma, \delta, q, F)$

$Q = \{q_0, q_1, q_2\}$
$\Sigma = \{a, b\}$
$q = \{q_0\}$
$F = \{q_2\}$

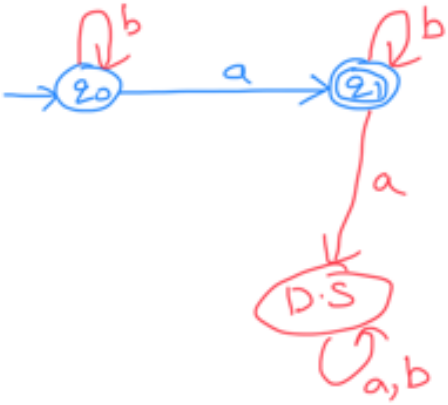
$$\left.\begin{aligned}
\delta(q_0, a) &= \{q_1\} \\
\delta(q_0, b) &= \{q_0\} \\
\delta(q_1, a) &= \{q_1\} \\
\delta(q_1, b) &= \{q_2\} \\
\delta(q_2, a) &= \{q_2\} \\
\delta(q_2, b) &= \{q_2\}
\end{aligned}\right\} \text{Transition function}$$

	a	b
q_0	$\{q_1\}$	$\{q_0\}$
q_1	$\{q_1\}$	$\{q_2\}$
q_2	$\{q_2\}$	$\{q_2\}$

Transition Table

Question-7- Design a minimal state DFA in which every string contains one 'a'. $\Sigma = \{a, b\}$

{ a, ab, ba, abb, bab, bba, bab ..... }



Correct strings [bbbbabbb, abbb, bbbba] → should be accepted

Incorrect string [bbbbbbb, bbbaabbb, ababbb] ⇒ should not accepted

$M(Q, \Sigma, \delta, q, F)$

$Q = \{q_0, q_1, DS\}$
$\Sigma = \{a, b\}$
$q = \{q_0\}$
$F = \{q_1\}$

$\delta(q_0, a) = \{q_1\}$
$\delta(q_0, b) = \{q_0\}$
$\delta(q_1, a) = \{DS\}$
$\delta(q_1, b) = \{q_1\}$

Transition Function

$\delta(DS, a) = \{DS\}$
$\delta(DS, b) = \{DS\}$

Transition Table

	a	b
q_0	$\{q_1\}$	$\{q_0\}$
q_1	$\{DS\}$	$\{q_1\}$
DS	$\{DS\}$	$\{DS\}$

Question-8- Design a minimal state DFA in which every string contains atleast one a.
$\Sigma = \{a, b\}$

$\{a, aa, ab, ba, aaa, aab, aba, baa, abb, bab, bba \ldots\ldots\}$

Correct string [abbbb, bbbba, ababab, bbababab] ⇒ Should accepted

Incorrect string [b, bb, bbb, bbbb, ⋮] ⇒ Not accepted

$M(Q, \Sigma, \delta, q, F)$

$Q = \{q_0, q_1\}$
$\Sigma = \{a, b\}$
$q = \{q_0\}$
$F = \{q_1\}$

$\delta(q_0, a) = \{q_1\}$
$\delta(q_0, b) = \{q_0\}$
$\delta(q_1, a) = \{q_1\}$
$\delta(q_1, b) = \{q_1\}$

Transition functions

Transition Table

	a	b
q_0	$\{q_1\}$	$\{q_0\}$
q_1	$\{q_1\}$	$\{q_1\}$

Question-9- Design a minimal state DFA in which every string contains atmost one a. $\Sigma = \{a, b\}$.

$\{\epsilon, a, b, ab, ba, bb, abb, bab, bba, bbb \cdots\}$

Question-10- Design a minimal state DFA in which every string contains odd no of a's. $\Sigma = \{a, b\}$.

{ a, ab, ba, aaa, abb, bab, bba, aaab ..... }

$n_a(w) \bmod 2 = 1$] odd no. of a's

$n_a(w) \bmod 2 = 0$] even no. of a's

Correct strings [ababaaba, bbaabab, aaabbb] ⇒ Should be accepted

Incorrect string [bbbbb, ababbb, bababaabb] ⇒ Should not accepted

$M(Q, \Sigma, \delta, q, F)$

$Q = \{q_0, q_1\}$
$\Sigma = \{a, b\}$
$q = \{q_0\}$
$F = \{q_1\}$

$$\left.\begin{aligned} \delta(q_0, a) &= \{q_1\} \\ \delta(q_0, b) &= \{q_0\} \\ \delta(q_1, a) &= \{q_0\} \\ \delta(q_1, b) &= \{q_1\} \end{aligned}\right\} \text{Transition function}$$

Transition Table

	a	b
$\rightarrow q_0$	$\{q_1\}$	$\{q_0\}$
$* q_1$	$\{q_0\}$	$\{q_1\}$

Question-11- Design a minimal state DFA in which every string contains even no. of a's $\Sigma = \{a, b\}$.

$\{\epsilon, b, aa, bb, aab, baa, bbb \text{ - - - -}\}$

$n_a(w) \bmod 2 = 0 \Rightarrow$ even no. of a's

$n_a(w) \bmod 2 = 1 \Rightarrow$ odd no. of a's

$M(Q, \Sigma, \delta, q, F)$

$Q = \{q_0, q_1\}$

$\Sigma = \{a, b\}$

$q = \{q_0\}$

$F = \{q_0\}$

$\delta(q_0, a) = \{q_1\}$

$\delta(q_0, b) = \{q_0\}$

$\delta(q_1, a) = \{q_0\}$

$\delta(q_1, b) = \{q_1\}$

	a	b
q_0	$\{q_1\}$	$\{q_0\}$
q_1	$\{q_0\}$	$\{q_1\}$

Transition Table

correct strings [bbbbb, ababaa, abbbbbabbb] ⇒ accept

incorrect string [abbbbb, bbbabbb, ababab] ⇒ should not accept

shortcut

odd no. of a's

$n_a(w) \bmod 2 = 1$

no. of states

reminder → final state

0, 2 → even

1, 3 → odd

shortcut

even no. of a's

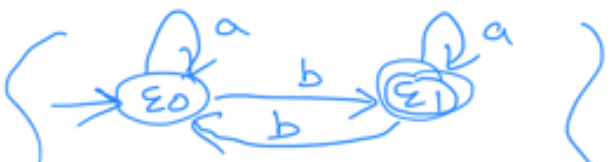
$n_a(w) \bmod 2 = 0$

0 ✓, 2 ✓, 4 ✓, 6, 8 → even

1, 3, 5 → odd

shortcut

$n_b(w) \bmod 2 = 1$ (odd no of b's)



0, 4, 6

1, 3, 5

Question-12- Design a minimal state DFA in which $n_a(w) \bmod 3 = 0$.

$\Sigma = \{a, b\}$

or

$n_a(w)$ is multiple of 3.

or

no. of a's mod 3 = 0.

correct strings ⇒

bbbbb ✓
babaabbbb ✓
bbbbaaab ✓
aaaaaa ✓

⇓

Should be accept

incorrect strings ⇒

abb ✗
baabbb
aabbaa

⇓

should not be accepted

$M(Q, \Sigma, \delta, q, F)$

$Q = \{q_0, q_1, q_2\}$
$\Sigma = \{a, b\}$
$q = \{q_0\}$
$F = \{q_0\}$

$\delta(q_0, a) = \{q_1\}$
$\delta(q_0, b) = \{q_0\}$
$\delta(q_1, a) = \{q_2\}$
$\delta(q_1, b) = \{q_1\}$

$\delta(q_2, a) = \{q_0\}$
$\delta(q_2, b) = \{q_2\}$

Transition Table

	a	b
$\rightarrow q_0$	$\{q_1\}$	$\{q_0\}$
q_1	$\{q_2\}$	$\{q_1\}$
q_2	$\{q_0\}$	$\{q_2\}$

Question-13- Design a minimal state DFA in which $n_b(w) \bmod 3 = 1$.

$M(Q, \Sigma, \delta, q, F)$

Transition Table

Question-14- Design a minimal state DFA in which $n_a(w) \bmod 3 = 2$. $\Sigma = \{a, b\}$.

$M(Q, \Sigma, \delta, q, F)$

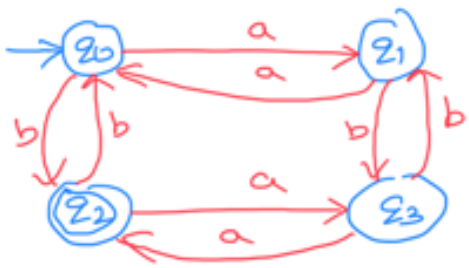
Transition Table

Question-15- Design a minimal state DFA in which every string contains even no. of a's and odd no. of b's. $\Sigma = \{a, b\}$

even no. of a's $\Rightarrow$ $n_a(w) \bmod 2 = 0$

and

odd no. of b's $\Rightarrow$ $n_b(w) \bmod 2 = 1$



$M(Q, \Sigma, \delta, q, F)$

Transition Table

Question-16- Design a minimal state DFA in which every string contains even no. of a's or even no. of b's.

$\Sigma = \{a, b\}$.

even no of a's $\Rightarrow$ $n_a(w) \bmod 2 = 0$

or

or

even no. of b's $\Rightarrow$ $n_b(w) \bmod 2 = 0$

$M\ (Q, \Sigma, \delta, q, F)$

Transition Table

Question-17- Design a minimal state DFA in which $\{ w \mid n_a(w) \bmod 3 = 1 \ \& \ n_b(w) \bmod 2 = 0 \}$

$\Sigma = \{a, b\}$

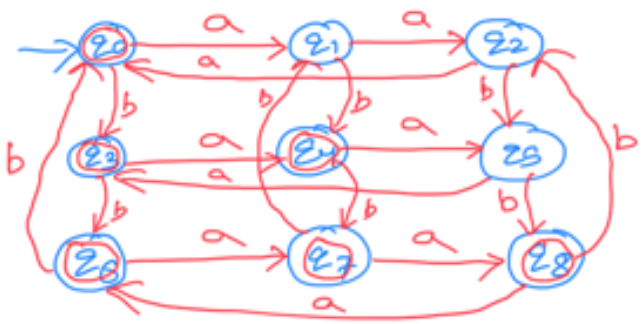
$M(Q, \Sigma, \delta, q, F)$

Transition Table

Question-18- Design a minimal state DFA in which

$$\{ w \mid n_a(w) \bmod 3 \leq n_b \bmod 3 \}$$

$$\Sigma = \{a, b\}$$



$$M(Q, \Sigma, \delta, q, F)$$

Transition Table

Question-19- Design a minimal state DFA which accepts string {100, 10, 1}
$\Sigma = \{0, 1\}$

{ 1, 10, 100}

$M(Q, \Sigma, \delta, q, F)$

$Q = \{q_0, q_1, q_2, q_3, DS\}$

$\Sigma = \{0, 1\}$

$q = \{q_0\}$

$F = \{q_1, q_2, q_3\}$

$\delta(q_0, 0) = \{DS\}$
$\delta(q_0, 1) = \{q_1\}$
$\delta(q_1, 0) = \{q_2\}$
$\delta(q_1, 1) = \{DS\}$
$\delta(q_2, 0) = \{q_3\}$
$\delta(q_2, 1) = \{DS\}$
$\delta(q_3, 0) = \{DS\}$
$\delta(q_3, 1) = \{DS\}$
$\delta(DS, 0) = \{DS\}$
$\delta(DS, 1) = \{DS\}$

Transition Table

	0	1
q_0	$\{DS\}$	$\{q_1\}$
q_1	$\{q_2\}$	$\{DS\}$
q_2	$\{q_3\}$	$\{DS\}$
q_3	$\{DS\}$	$\{DS\}$
DS	$\{DS\}$	$\{DS\}$

—

Question-20- Design a minimal state DFA in which w is string, $w \in \{a,b\}$, $|w| \leq 3$.

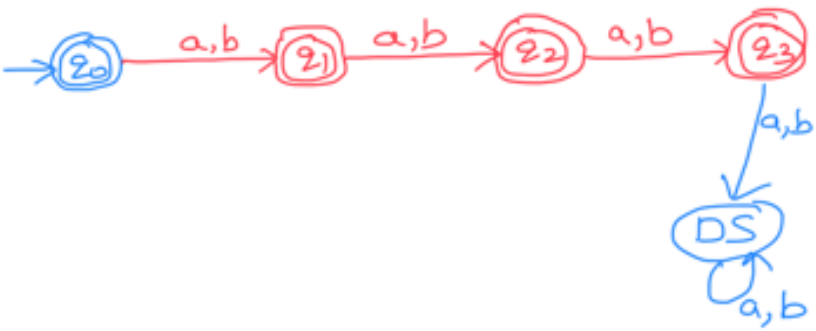
0 length string — ϵ (epsilon)

1 " " — $\{a, b\}$

2 " " — $\{aa, ab, ba, bb\}$

3 " " — $\{aaa, aab, aba, baa, abb, bab, bba, bbb\}$

$\{\epsilon, a, b, aa, ab, ba, bb, aaa, aab, aba, baa, abb, bab, bba, bbb\}$

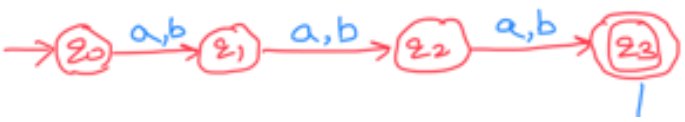


$M(Q, \Sigma, \delta, q, F)$

Transition Table

Question-21 – Design a minimal state DFA in which w is string, $w \in \{a,b\}$, $|w| = 3$

3 length string – {aaa, aab, aba, baa, abb, bab, bba, bbb}



$m(Q, \Sigma, \delta, q, F)$

Transition Table

Question-22- Design a minimal state DFA in which w is string, $w \in \{a,b\}$, $|w| \geq 3$

3 length — Allowed
4 length
5 length
6 len
⋮

$M(Q, \Sigma, \delta, q, F)$

Transition Table

Ques-23 – Design a minimal state DFA which contains strings over $\{0,1\}$, if strings interpreted as binary number then it should be divisible by 3.

$\Sigma = \{0,1\}$

$w = 110$

Binary no. $= \underset{2^2\ 2^1\ 2^0}{110} \Rightarrow 1\times2^2 + 1\times2^1 + 0\times2^0$

$4 + 2 + 0 = 6$

Decimal no. $= 6$

Should be divisible by 3

$\Downarrow$

yes accepted

$w = 111$

$1\times2^2 + 1\times2^1 + 1\times2^0 = 4+2+1 = 7$

Div by 3

No, not accepted

Transition Table

$\Sigma = \{0,1\}$

No. of states = 3

	0	1
$\rightarrow *q_0$	$\{q_0\}$	$\{q_1\}$
q_1	$\{q_2\}$	$\{q_0\}$
q_2	$\{q_1\}$	$\{q_2\}$

$M(Q, \Sigma, \delta, q, F)$

Ques-24- Design a minimal state DFA which contains

Ques-21 Design a minimal state DFA which contains strings over $\{0,1,2\}$ if strings interpreted as ternary number then it should be divisible by 4.

$w = 22$

$\hookrightarrow 2\times3^1 + 2\times3^0 = 6+2 = 8$

Div by 4

Yes, Accepted

$w = 21$

$\hookrightarrow 2\times3^1 + 1\times3^0 = 6+1 = 7$

DIV by 4

No

Not Accepted

Transition Table

$\Sigma = \{0,1,2\}$

No. of states = 4

	0	1	2
$\rightarrow$ * q_0	$\{q_0\}$	$\{q_1\}$	$\{q_2\}$
q_1	$\{q_3\}$	$\{q_0\}$	$\{q_1\}$
q_2	$\{q_2\}$	$\{q_3\}$	$\{q_0\}$
q_3	$\{q_1\}$	$\{q_2\}$	$\{q_3\}$

22

Accepted

$\{ \}$

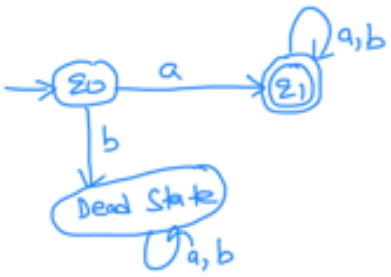
Not accepted

$M(Q, \Sigma, \delta, q, F)$

Ques-25- Design a minimal state DFA in which strings doesn't starts with 'a'.
$\Sigma = \{a, b\}$.

Starts with a

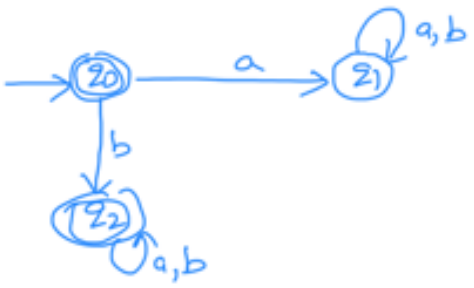
$\{a, aa, ab, aaa, aba, abb \ldots\ldots\}$



q_0
q_1
DS

Not starts with 'a'

Do complement of starts with 'a' (Final → Non final, Non final – final)



ab – No

bab – Yes

$M(Q, \Sigma, \delta, q, F)$

Transition Table

Ques-26- Design a minimal state DFA in which strings doesn't contain the substring 'ab'
$\Sigma = \{a, b\}$

Substring ab

Not contain substring ab

Do complement

bbba – Yes

baba – No

$M(Q, \Sigma, \delta, q, F)$

Transition Table